

Department of Mechanical Engineering
University of South Carolina
Columbia, SC 29208
January 30, 2026

Dear Members of the Evaluation Committee:

I recommend Samuel Cancilla for the SC Space Grant Undergraduate Research Award. I have supervised Samuel in the Integrated Multiphysics and Systems Engineering Laboratory (iMSEL) since March 2025, and he is among the most capable undergraduates I have worked with in my eight years at the University of South Carolina.

What sets Samuel apart is that he owns problems end-to-end. When he joined iMSEL, I gave him the open-ended task of developing a radar-based tracker for autonomous maritime perception. Rather than waiting for a scripted plan, he scoped the data requirements himself, organized multi-site radar collection campaigns across three South Carolina lakes and Charleston Harbor—over 38 hours total—and built the full detection-and-tracking pipeline from scratch. During one campaign at Lake Greenwood, unexpected wind gusts changed the clutter environment mid-collection; he revised the collection grid on-site to capture both regimes, turning a setback into a richer dataset. That kind of initiative is rare at any level, let alone from an undergraduate.

His technical judgment is equally strong. He chose to implement the tracking pipeline in Rust for an embedded NVIDIA Jetson Orin platform, and when profiling revealed memory-allocation bottlenecks in the clustering loop, he independently redesigned the pipeline around pre-allocated buffers before raising it at our group meeting—arriving with a working solution, not just a problem. He is now co-author on a paper submitted to AIAA, and the algorithm presented in that paper is entirely his work. In my experience, most undergraduates contribute to existing codebases; Samuel created one.

His proposed research—extending the adaptive clustering to temporal self-tuning and adding a vessel classifier—is well scoped. He completed the current pipeline from algorithm design through embedded deployment and ablation testing in 10 weeks; the proposed extensions are narrower, and the 14-week timeline is realistic. Samuel himself identified the NASA connection during a lab presentation, drawing the parallel between GPS-denied maritime perception and analogous challenges in lunar and planetary surface missions. I did not prompt that insight.

Beyond the lab, Samuel serves as an Army National Guard 09R SMP Cadet through ROTC and as Assistant Editor of The Daily Gamecock. He maintains a 3.5 GPA while balancing military obligations, editorial deadlines, and research fieldwork. The operational discipline he brings—equipment checklists, detailed field logs, post-campaign debriefs—is something I have come to rely on when planning our lab’s data collection schedule.

I give Samuel my strongest recommendation for this award.

Sincerely,

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